

AIPL Runtime Feature Matrix

7 implementations compared

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Overview

AIPL (Actor-based Intelligent Parallel Language, formerly AIPL) has grown into a **seven-runtime project**. This document provides a side-by-side comparison of every language and infrastructure feature across all implementations, grounded in source surveys.

Legend. ✓ = full support, △ = partial, ✗ = not supported, – = not applicable.

	Short name	Source
The seven runtimes.	Python (annotated)	src/python-aipl/
	Python (inferred)	src/python-aipl-inferred/
	OCaml	src/*.ml, _build/.../repl_thread.exe
	JS-OCaml (server)	src/app.js, console_server.js, ide.js + OCaml web_gateway
	JS-Browser (serverless)	src/browser-abcl/
	JS-Node (server)	src/node-aipl-server/ (Node HTTP server wrapping browser-abcl)
	C (aipl2c)	src/aipl2c.ml codegen + src/abcl_gui_runtime.c

Column abbreviations in matrices below. **Py-A** = Python annotated, **Py-I** = Python inferred, **OCaml**, **JS-O** = JS via OCaml backend, **JS-B** = JS in browser, **JS-N** = JS in Node server, **C** = aipl2c codegen.

Inheritance relationships.

- **JS-OCaml** is a thin HTTP client over the OCaml backend, so its feature set inherits OCaml's (plus the new `/api/typecheck` endpoint).
- **JS-Node** wraps the same browser-abcl parser / type-checker / interpreter inside a Node.js HTTP server. Runtime feature set therefore mirrors JS-Browser.
- **Python (inferred)** imports parser, interpreter, AI, remote, and dashboard from `../python-aipl/`. All *runtime* features match Python (annotated); only the *static type-checker* layer differs (full Hindley-Milner instead of Phase 11+ annotation-driven).

Sample programs

Number of `.abcl` sample programs reachable by each runtime (core directory + AI integration + remote-actor demos):

	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
core samples	33	33	57	57	5	5	57
AI integration samples	11	11	8	8	0	0	8
remote-actor samples	8	8	1	1	0	0	1
total	52	52	66	66	5	5	66

Cross-cutting / not tied to a specific runtime:

- `aipl-self-host/`: 48 AIPL-in-AIPL self-host programs
- `docker/cross/samples/`: 4 cross-language interop samples

Note that the C runtime processes the same `.abcl` files as OCaml via `aipl2c`; of the 66 reachable samples, 48 currently pass the `aipl2c` smoke test (the remaining 9 in `abclc/` have pre-existing type ambiguities surfaced by our hard-fail policy).

What each sample exercises

Samples are grouped by the language or infrastructure feature they demonstrate. “Where” points to the source directory; “Target” lists the runtime(s) and codegen targets that actually execute the sample.

Target abbreviations: **Py** = python-aipl / python-aipl-inferred; **OCaml** = OCaml REPL (`abclc`); **C** = `aipl2c` → C+pthread; **SDL2** = `aipl2c` → C+SDL2 GUI binary; **Xinu** = `aipl2c --xinu` → Xinu-flavoured C; **Py-gen** = `aipl2c --python` → stand-alone Python; **Pony** = `aipl2c --pony` → Pony actor source; **Erlang** = `aipl2c --erlang` → Erlang module; **Go** = `aipl2c --go` → Go source; **Prolog** = `aipl2c --prolog` → SWI-Prolog `.pl`; **JS-B** = browser-abcl; **JS-N** = node-aipl-server.

Feature category	Sample(s)	Where	What it checks	Target
Basic actor model now/future/await	Hello, Counter, PingPong NowFuture, CooperativeNowFuture, NowFutureDemo	py-aipl + abclc py-aipl + samples-ai + abclc/ai-samples	actor creation, send , field state, self-send three message-passing forms in one program	Py, OCaml, C, JS-B, JS-N Py, OCaml, C, JS-B, JS-N
Bounded mailboxes / select	BoundedBuffer, bounded_buffer{,_visual}	py-aipl + abclc + browser-abcl	producer/consumer, selective receive, visualisation	Py, OCaml, C, JS-B, JS-N
Dining philosophers	Philosophers, philosophers{-1,-2}, Philosophers5{,_debug,_trace,Gui,Py,Xinu}	py-aipl + abclc + browser-abcl	fork as actor, deadlock-free serialisation, SDL/Python/Xinu variants	Py, OCaml, C, SDL2, Py-gen, Xinu, JS-B, JS-N
become (class swap)	become, bbecome	abclc	runtime actor-class replacement	Py, OCaml
select / selective recv	Channels, Channels2	py-aipl/samples	typed channels, worker pool with request/result	Py
Method injection	MethodPatch	py-aipl/samples	add_method/remove_method runtime patching	Py

Feature category	Sample(s)	Where	What it checks	Target
Dynamic compile	Dynamic, DynamicWorkerPool	py-aipl/samples	<code>compile()</code> builtin → runtime class generation	Py
Top-level functions	Functions, Signatures	py-aipl/samples	user functions, overload signatures, <code>typeof</code>	Py
Records	Records	py-aipl/samples	<code>{a:int, b:string}</code> literal, field access, structural <code>typeof</code>	Py
Tuples	Tuples	py-aipl/samples	positional, immutable, mixed slot types, nesting	Py
Arrays	Arrays, MultiDimArrays	py-aipl/samples	static-sized, multi-dim, dynamic sizes	Py
Gradual type checker	Typecheck, Typecheck11{b,c,de}	py-aipl/samples	Phase 11 → 11e: literals, call-site validation, unions/generics, narrowing, length-tagged arrays	Py
Phase 11 typed counter	Phase11_TypedCounter	abclc	typed annotations / generics / <code>typeof</code> narrowing	OCaml
Phase 12 effects	Effects, Phase12_EffectsLog	py-aipl + abclc	capability-based <code>!{fs,ai,net,mut}</code> system	Py (static), OCaml (sample)
Phase 13 channels	Phase13_Channels	abclc	CSP channels (design doc + runtime in Py)	Py (runtime), OCaml (sample)
Phase 14 linear types	Linear, Linear2, Phase14_Linear	py-aipl + abclc	use-after-move; DB transaction with linear handle	Py (static), OCaml (sample)
Phase 15 owned fields	Owned, Owned_violations, Phase15_Owned	py-aipl + abclc	<code>pub</code> field visibility, intentional violations	Py (static), OCaml (sample)
Phase 16 transient cast	Transient, Transient_violation	py-aipl/samples	runtime type cast at any-boundary	Py
Phase 17 structured conc.	Phase17_StructuredConc	py-aipl/samples	<code>scope { future ... }</code> auto-joining	Py
Session types / protocol traces	SessionTyped	py-aipl/samples-ai	runtime-checked typed session protocols (arg + reply types); order-only protocol traces; AIOS service registry	Py
AI integration (mock + real)	AIActor, AIChain, AIChainReal, AIHello, MultiProvider	py-aipl + samples-ai + abclc/ai-samples	LLM-backed actors; multi-provider; chained pipeline	Py, OCaml
AI governance	Budgeted	py-aipl/samples-ai + abclc/ai-samples	token budget, concurrency cap, fallback chain	Py, OCaml
AI cooperative pattern	CooperativeNowFuture{-jp,-jp-remote}, CooperativeSolve{-jp,Remote,Remote-jp}, Reviewer, Fanout, PriorityFanout	py-aipl/samples-ai + abclc/ai-samples	Planner→Solver→Reviewer; fan-out aggregator; priority routing	Py, OCaml

Feature category	Sample(s)	Where	What it checks	Target
Remote actors	client, coordinator, verifier, RemoteCalcClient/Server	server, solver, reviewer_node*,	py-aipl/samples-remote + abclc/samples-remote + abclc/ai-samples	HTTP cross-machine sends; HMAC-signed coordination
Web / dashboard	web_calc, SiteGen	abclc + py-aipl/samples	embedded HTTP gateway; static-site generator	Py (SiteGen), OCaml (web_calc)
GUI / SDL2	Rotate{One,Three,Four}Lines{,Gui}, MultiLineSpin, Philosophers5Gui, BoundedBufferGui, DisasterReturnGui, LineDrawer, window	abclc	SDL2-backed GUI codegen via aipl2c	SDL2 (via aipl2c)
Python codegen target	BoundedBufferPy, Philosophers5Py, Rotate4LinesPy	abclc	aipl2c --python emits stand-alone Python	Py-gen
Xinu (embedded OS) target	BoundedBufferXinu, Philosophers5Xinu, Rotate4LinesXinu	abclc	aipl2c --xinu emits Xinu-flavoured C	Xinu
Pony codegen target	Hello, counter (verified); PingPong (xfail)	abclc	aipl2c --pony emits Pony source; class → actor , methods → be ; two-step _aipl_init decouples construction from init body	Pony
Erlang codegen target	Hello, counter (verified); PingPong (xfail)	abclc	aipl2c --erlang emits an .erl module; class → spawn + receive loop; fields → loop args with versioned variables on assign	Erlang
Go codegen target	Hello, counter (verified); PingPong (xfail)	abclc	aipl2c --go emits a single main.go ; class → struct + goroutine; methods → typed message structs + helper methods + type-switch dispatch	Go
Prolog codegen target	Hello, counter (verified); PingPong (xfail)	abclc	aipl2c --prolog emits a single SWI-Prolog .pl using library(thread) ; class → c_loop(Fields) thread + thread_get_message + Msg = m(Args) → body dispatch; expressions hoisted to goals (X is A + B , format(atom(S), ...))	Prolog
Drone / simulation	drone_simulator	browser-abcl	obstacle-aware drone swarm with comm + view range	JS-B, JS-N
Trace / minimal repros	H, P, T*, LD*, MS, AA, PP, PH, line*, Philosophers5_{debug,trace}	abclc	reduced repro cases used during runtime / TLA+ / Spin model-checking	OCaml

Core language

Feature	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
method injection (<code>add_method/remove_method</code>)	✓	✓	×	×	×	×	×
<code>now</code> synchronous send	✓	✓	✓	✓	✓	✓	✓
<code>future</code> async send	✓	✓	✓	✓	✓	✓	✓
<code>await</code> future block	✓	✓	✓	✓	✓	✓	✓
<code>send</code> fire-and-forget	✓	✓	✓	✓	✓	✓	✓
<code>become</code> actor class swap	✓	✓	✓	✓	×	×	×
<code>select</code> selective receive	✓	✓	✓	✓	✓	✓	×
top-level functions	✓	✓	△	△	×	×	✓
signatures / overloads	×	×	✓	✓	×	×	×
dynamic compile (<code>compile()</code>)	✓	✓	×	×	×	×	×
worker pool / <code>DynamicWorkerPool</code>	✓	✓	×	×	△	△	×

Type system

Feature	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
type inference	△trace	✓HM	✓HM	✓HM	✓flow	✓flow	✓HM+spec
type annotations	✓	△ignored	×	×	×	×	×
records <code>{a:int, b:str}</code>	✓	✓	△type only	×	×	×	△type only
tuples <code>(1, "a")</code>	✓	✓	×	×	×	×	×
arrays (typed, multi-dim)	✓	✓	✓	✓	△	△	△
generics on functions	×	✓Forall	△Forall	△	×	×	×

Phase 11+ advanced features

Python (inferred) runs HM only, so the annotation-driven Phase 11+ static checks that Python (annotated) performs are not re-implemented. Programs still execute at runtime via the shared interpreter, but the static guarantees from Phase 12 / 14 / 15 / 16 are lost. Phase 13 (channels), Phase 17 (**scope**), and the session-type / protocol-trace / AIOS-registry families are runtime features, so they remain available in Py-I.

Session types here are *runtime-checked* and explicitly separate from HM inference: they observe values flowing over the wire at runtime and validate them against a declared protocol spec (`"actor.method(arg_t1, ...) ! ret_type -> ..."`). Violations are recorded into `session_events()` and counted; the program does not abort.

Feature	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
channels (CSP-style) —Phase 13	✓	✓	×	×	×	×	×
linear types / use-after-move —Phase 14	✓	×	×	×	×	×	×
owned / pub fields —Phase 15	✓	×	×	×	×	×	×
effects <code>!{fs,ai,net,mut}</code> —Phase 12	✓	×	×	×	×	×	×
structured concurrency <code>scope</code> —Phase 17	✓	✓	×	×	×	×	×
transient cast at any-boundary —Phase 16	✓	×	×	×	×	×	×
session types (runtime-checked typed protocols)	✓	✓	×	×	×	×	×
protocol traces (order-only)	✓	✓	×	×	×	×	×
AIOS service registry (alias-resolved send)	✓	✓	×	×	×	×	×

Networking, AI, infrastructure

Feature	Py-A	Py-I	OCaml	JS-O	JS-B	JS-N	C
remote actors (HTTP)	✓	✓	△no WS	△	×	△srv only	✓via OCaml
WebSocket	✓ws_listen (web-sock-ets lib)	✓(send/recv in A)	△no WS (verified 101)	✓via OCaml	✓browser native	✓/ws?sid=abcl_ws_runtime.c + /api/broadcast (ws lib)	✓+ lib-sock-ets
AI integration (<code>ai_call</code>)	✓streaming	✓	✓	✓	✓mock	✓mock	✓
provider arg (<code>ai_call([1 2 3,] prompt)</code>) — 1=gemini, 2=anthropic, 3=openai	✓	✓	✓	✓	✓	✓	✓(via OCaml)
<code>now actor.m(...)</code> + <code>ai_call</code> inside method (blocking)	✓	✓	✓	✓	✓	✓	×
<code>future actor.m(...)</code> + <code>await(f)</code> + <code>ai_call</code>	✓	✓	✓	✓	✓	✓	×
<code>send</code> + callback pattern with <code>ai_call</code>	✓	✓	✓	✓	✓	✓	×
AI governance (budget/concurrent/fallback)	✓	✓	✓	✓	×	×	✓
HMAC-signed remote send	✓	✓	×	×	×	×	✓
persistent actor state	✓	✓	×	×	×	×	×
live dashboard (SSE)	✓	✓	△polling	△	×	×	✓
<code>/api/typecheck</code> JSON endpoint	×	×	✓	✓	×	✓	—
<code>/api/run</code> JSON endpoint	×	×	×	×	×	✓	—

Actor concurrency model

Every AIPL implementation runs each actor as a separate concurrent unit, but the kind of unit differs by 3–4 orders of magnitude in weight. This affects how many actors a program can usefully spawn, how messages interleave, and whether blocking I/O in one actor stalls the whole runtime.

Runtime target	codegen	Model	Per-actor unit	Source pointer
Python (annotated)		1:1 OS thread	<code>threading.Thread(daemon=True)</code>	<code>aipl_runtime.py:92</code>
Python (inferred)		1:1 OS thread	(inherits from python-aipl)	(re-uses interp)
OCaml		1:1 OS thread	<code>Thread.create</code>	<code>eval_thread.ml:~1168</code>

Runtime / codegen target	Model	Per-actor unit	Source pointer
JS-OCaml (server)	1:1 OS thread	(= OCaml backend threads)	(= OCaml)
JS-Browser (browser-abcl)	cooperative event loop	setTimeout-driven mailbox drain	runtime.js:~233
JS-Node (node-aipl-server)	cooperative event loop	(= browser-abcl runtime)	re-export
C (default)	1:1 OS thread	pthread_create	c_translator.ml:~721,790
C + SDL2	1:1 OS thread	pthread_create	(same as default C)
C + Xinu	1:1 OS process (kernel-level)	Xinu create() + ready()	c_translator.ml:~1077
C → Python codegen	1:1 OS thread	threading.Thread (o._spawn())	c_translator.ml:~1847
C → Pony codegen	M:N lightweight (runtime-scheduled)	Pony actor (work-stealing)	implicit in generated actor
C → Erlang codegen	M:N lightweight (BEAM)	BEAM (spawn(fun()->...end)) process	c_translator.ml:~2413
C → Go codegen	M:N lightweight (Go runtime)	goroutine (go c.run())	c_translator.ml:~2728
C → Prolog codegen	1:1 OS thread	SWI thread (thread_create/3)	c_translator.ml:~3060

Three concurrency tiers

- **1:1 OS thread / process** (Python, OCaml, C, C-SDL2, Xinu, C→Python, C→Prolog): each actor is a kernel-scheduled thread. Simple mental model; blocking I/O in one actor doesn't stall others. Scales to ~100–1000 actors before kernel overhead dominates.
- **M:N lightweight** (Pony, Erlang, Go): actors are scheduled by the language runtime onto a small pool of OS threads. Scales to $\sim 10^5$ – 10^7 actors. Blocking syscalls in one actor may stall its scheduler unless the runtime intercepts them (Erlang's BEAM intercepts; Go's runtime intercepts via `netpoll`; Pony's runtime requires non-blocking idioms).
- **Cooperative event loop** (browser-abcl, node-aipl-server): single OS thread; mailboxes are drained turn-by-turn through `setTimeout(0)` re-entry into the event loop. No real parallelism, but message-level interleaving is preserved. A long synchronous computation in one actor *will* stall all others — by design, since this is the only concurrency model the browser DOM allows.

Cross-cutting implication

The AIPL programming model (`send` / `now` / `future` / `await`) is preserved across all three tiers — the choice of target picks a point on the cost/scale curve without changing the program text. The same `.abcl` file:

- runs ~1000 actors fine on Python / OCaml / C (OS-thread tier);
- scales to millions of actors on Erlang / Go / Pony codegen;
- runs in a browser sandbox via JS-Browser at the cost of single-threaded execution.

C-version-specific codegen targets

Target	C
C + pthread standalone binary	✓
C + SDL2 GUI binary (1178-line runtime)	✓
Xinu embedded-OS target (<code>--xinu</code>)	✓
Python target (<code>--python</code>)	✓
Pony target (<code>--pony</code>) —class→actor, methods→be, two-step <code>_aipl_init</code>	✓
Erlang target (<code>--erlang</code>) —class→spawn+receive loop, fields→versioned loop args	✓
Go target (<code>--go</code>) —class→struct+goroutine, methods→typed message structs + type-switch dispatch	✓
Prolog target (<code>--prolog</code>) —class→SWI thread + receive loop, expressions hoisted to goals (<code>is/2</code> , <code>format(atom(...))</code>)	✓

Key observations

Python (annotated) is the most feature-complete runtime

Of the 30 surveyed features, Python (annotated) ticks roughly 28. It owns the full Phase 11–17 stack (channels, linear types, owned fields, effects, structured concurrency, transient casts), method injection, dynamic compile, persistent actor state, and full AI integration (including streaming and images).

Python (inferred) is Python’s HM-typed twin

Runtime features match Python (annotated) exactly because the interpreter is shared. The trade is on the static side: the Phase 11+ annotation-driven checks are replaced with a Hindley-Milner inference pass. Inferred types cross-verified against OCaml: on shared samples (`Hello.abcl`, `counter.abcl`) the two runtimes produce identical type schemes.

OCaml is the canonical core

Implements roughly 15 of the surveyed features. Phase 11+ exist only as `.abcl` design-document samples, not implemented. Solid `become` / `select` / HM inference; remote is HTTP only (no HMAC, no WebSocket).

JS-OCaml $\approx$ OCaml

Thin HTTP client over the OCaml backend, so features inherit. The only differentiator is the `/api/typecheck` endpoint added in commit 5227f87.

JS-Browser is the minimal implementation

Around 8 of the 30 features. Basic actor model plus our newly-added flow-sensitive type inference. No `become`, no channels, no remote. AI integration exists but is mock-only.

JS-Node (Node-server, newest)

Wraps the browser-abcl parser, type-checker, and interpreter inside a Node.js HTTP server. Exposes `/api/typecheck` (the same JSON shape as OCaml’s endpoint) and `/api/run` (executes and returns stdout) —the only runtime with the latter. Runtime feature set mirrors browser-abcl; the value-add is having both the type-checker and the interpreter reachable over HTTP without bringing up either OCaml or a browser.

C version — surprisingly strong on infrastructure

About 17 of 30. `become` and `select` are not implemented (codegen emits `/* unsupported */`), but it inherits OCaml’s HMAC, SSE, AI governance, and adds **three additional codegen targets** (SDL2, Xinu, Python). Fields, parameters, and locals are specialized to native C types.

Method injection — a Python-family exclusive

Both Python variants use mutable method dispatch tables that can be mutated at runtime via `add_method` / `remove_method`. Other runtimes substitute `become` (whole-class swap) as the closest equivalent. `JS-Browser` and `JS-Node` lack even `become` (pure actor execution).

Type inference cross-verification

For shared samples (`abclc/Hello.abcl`, `abclc/counter.abcl`), the three HM-based runtimes (**OCaml**, **Python-inferred**, **C**) produce identical inferred types:

```
Hello:  count : float
        init  : (int) -> unit
        greet : () -> unit
        inc   : () -> unit
```

```
Counter: count : float
        inc   : () -> unit
        dec   : (float) -> unit    <- monomorphized via call site
```

The flow-sensitive variants (**JS-Browser**, **JS-Node**) reach the same answers on shared samples within the limits of their algorithm —field types match, but method signature schemes are not produced the same way.

Phase 11+ implementation gap

The full Phase 11–17 stack is implemented only in Python (annotated). This represents AIPL’s language-evolution frontier; Python (annotated) serves as the research runtime. Python (inferred) keeps the runtime side but trades the static checks for HM-style polymorphic inference — a different research axis (declarative typing vs. annotated capabilities).

Generated 2026-05-13. Includes `python-aipl-inferred` (commit `270f291`) and `node-aipl-server` (commit `adeb0b6`). For source pointers, run `grep` against the files listed in each runtime’s source column.